

SANTOSH PANT

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EDUCATION

Knox College

Sept 2023 - June 2027

Bachelor of Science in Computer Science and Data Science (GPA: 3.82)

Galesburg, Illinois

- **Honors:** Break Through Tech AI Fellow (Cornell Tech, 2025-26); Richter Research Scholar; Dean's List (6/6 terms); ICPC North Central Participant (UIUC site, 2025); Teaching Assistant - Data Structures & Algorithms; CS Club Lead
- **Hackathons:** UMich AI for Heat Resilience - 2nd Place (\$2K); HackIllinois 2026 (UIUC) - Best Use of Action VectorAI DB (1st, sponsor track); ASA DataFest 2026 - Best Insights Track; HackAugie 2026 - Best Gaming Hack

GEOSPATIAL ML RESEARCH EXPERIENCE

Machine Learning Research Collaborator

June 2026 - Present

RGB-to-Thermal Urban Heat Mapping - Mentor: Prof. Geoffrey Siwo, UMich Center for Global Health Equity

Remote

- Continuing the 2nd-place UMich AI for Heat Resilience project with the event's organizing professor: predicting street-level urban heat maps from RGB drone imagery over Ann Arbor, toward cheap, scalable heat-risk mapping without thermal cameras; targeting a publication (github.com/Santoshpant23/rgb-to-thermal).
- Diagnosed two data faults capping accuracy (false-color palette targets instead of radiometric temperature; unaligned RGB/thermal camera pairs - alignment alone moved PSNR 16.9 to 18.4 dB); ensembled CNN, GAN, and physically-structured models to reach 19.3 dB PSNR / 0.71 SSIM on a 202-image held-out test set (constant-prediction baseline: 10.7 dB).

Satellite Heat-Stress Forecasting, Nepal Terai

2026 - Present

Mentor: Prof. Andrew Leahy, Knox College (first-author paper under review at ACM SIGSPATIAL 2026)

Galesburg, Illinois

- Built an early-season vegetation heat-stress pipeline over 21 Terai rice districts using Google Earth Engine (MODIS LST/NDVI, ERA5-Land, ESA WorldCover; 79,782 pixel-years); deployed a live forecast app (terai-heat-forecaster.streamlit.app).
- Showed previous-year persistence outperforms fold-safe XGBoost on average (macro-F1 0.476 vs 0.362) under year-blocked leave-one-district-out evaluation with paired district-block bootstrap (N=5,000); proposed persistence plus modal-class nulls as required baselines.

scaleshift-bench: Geospatial Foundation-Model Benchmark

2026 - Present

Independent researcher, Knox College (in prep)

Galesburg, Illinois

- Benchmarked four Earth-observation foundation models (Clay v1.5, Prithvi-EO-2.0, TerraMind, AnySat) on cropland and tree-cover mapping across Nepal, India, and Mozambique; showed evaluation protocol (per-polygon vs. per-pixel) reorders rankings and the field-size gradient is model-agnostic.

Smallholder Burn-Detection Audit, Nepal Terai

2026 - Present

Independent researcher, Knox College

Galesburg, Illinois

- Audited MODIS MCD64A1, VIIRS VNP14A1, and Sentinel-2 BAIS2 burn detection against 250+ PlanetScope 3 m reference chips; showed the BAIS2 index over-detects burned area by roughly three orders of magnitude and identified phenology-driven over-detection and cropland-mask leakage as systematic failure modes.

ENGINEERING EXPERIENCE

Software Engineering Intern - The MACRO Consortium

March 2025 - June 2025

- Shipped a priority-based scheduling algorithm into the open-source pyscope astronomy package; investigated and mitigated a data-loss bug, improving reliability for nightly observation batches at Knox College and partner institutions.

SELECTED PROJECTS

CropScan | PyTorch (EfficientNet-B0, MobileNetV2), FastAPI, React, MongoDB, Docker

cropscan.tech

- Built and deployed a full-stack crop-disease diagnosis app for smallholder farmers: dual-CNN leaf-image classification plus AI-generated treatment guidance, shipped with JWT auth, rate limiting, Docker Compose, and CI via GitHub Actions.

SKILLS

- **Geospatial & Remote Sensing:** Google Earth Engine, GIS, Rasterio, geopandas, Sentinel-1/2, MODIS, Landsat, PlanetScope, ERA5, ESA WorldCover/WorldCereal, AlphaEarth embeddings, drone imagery
- **ML & AI:** PyTorch, deep learning, computer vision, scikit-learn, XGBoost, Diffusers, EO foundation models (Clay, Prithvi, TerraMind, AnySat), CNNs (ResNet, U-Net), Grad-CAM
- **Languages:** Python, R, TypeScript, Go, SQL
- **Tools:** FastAPI, Streamlit, Folium, PostgreSQL, AWS (cloud), Docker, Git, Linux, data pipelines